

Linux Class 3 - Task

Searching and Editing in the vi Editor

For practice I made a folder called class3 and inside it a file called practice.txt with six lines. I opened it by typing `vi practice.txt`. I also typed `:set number first` so that line numbers are shown, because it makes the changes easier to see in the screenshots.

One thing I learnt is that in vi I must press the Esc key first every time. Esc puts vi back in command mode. If I forget, whatever I type just goes into the file as normal text.

1. How to search and replace a case insensitive pattern in vi editor?

I press Esc and then type this command at the bottom of the screen:

```
:%s/linux/UNIX/gi
```

Each part of the command means something:

- `:` starts a command at the bottom of the screen.
- `%` means do it in the whole file, not only the line I am on.
- `s` means substitute, which is search and replace.
- `/linux/UNIX/` means find linux and put UNIX in its place.
- `g` means change every one on the line, not just the first one.
- `i` means ignore capital letters, so it is case insensitive.

The `i` at the end is the important one for this question. Without it, only the small letter word would change.



```
Shiva@DESKTOP-MRCVEKT: ~/class3
1 Linux is an operating system.
2 linux has many commands.
3 An error was found in the log file.
4 LINUX is open source software.
5 Another Error appeared today.
6 This is the last line.

:%s/linux/UNIX/gi
```

Screenshot 1: The file before, with the command typed at the bottom.

My file had Linux, linux and LINUX written in three different ways. After pressing Enter all three were changed and vi showed the message 3 substitutions on 3 lines at the bottom.



```
Shiva@DESKTOP-MRCVEKT: ~/class3
1 UNIX is an operating system.
2 UNIX has many commands.
3 An error was found in the log file.
4 UNIX is open source software.
5 Another Error appeared today.
6 This is the last line.

3 substitutions on 3 lines
```

Screenshot 2: After pressing Enter, all three spellings became UNIX.

Another way is to type :set ic first, which means ignore case. Then I can use :%s/linux/UNIX/g without the i.

2. How to search the pattern in forward direction?

I press Esc, then type a slash and the word I am looking for, and press Enter:

```
/UNIX
```

The cursor jumps down to the next place where the word is found. To keep going to the next one I press n. If I go too far, N takes me back one match. The word I searched for stays at the bottom of the screen and the numbers on the right show which line the cursor is on now.



Screenshot 3: Searching forward with /UNIX and then pressing n.

3. How to search the pattern in reverse direction?

It is the same thing but with a question mark instead of a slash:

```
?UNIX
```

Now vi searches upwards instead of downwards. Pressing n keeps moving up to the match before, and N goes the other way. In the screenshot the cursor moved up to line 2.



Screenshot 4: Searching backward with ?UNIX.

4. How to insert a blank line where the cursor is pointing in vi editor?

I press Esc first, then press capital O:

```
O
```

A blank line opens on the line the cursor was on, the old line moves down by one, and vi goes into insert mode. I can see -- INSERT -- at the bottom of the screen. If I only want the blank line I press Esc again without typing anything.

Small o does almost the same but it opens the blank line below instead of above.



```
Shiva@DESKTOP-MRCVEKT: ~/class3
1 UNIX is an operating system.
2 UNIX has many commands.
3
4 An error was found in the log file.
5 UNIX is open source software.
6 Another Error appeared today.
7 This is the last line.
-- INSERT -- 3,1 All
```

Screenshot 5: Pressing O put a blank line 3 and pushed the rest of the file down.

Commands I used

Command	What it does
:%s/old/new/gi	Search and replace in the whole file, ignoring capital letters.
:set ic	Makes all searches ignore capital and small letters.
/word	Search forward (downward) for the word.
?word	Search backward (upward) for the word.
n	Go to the next match in the same direction.
N	Go to the match in the opposite direction.
O	Insert a blank line above the line the cursor is on.
o	Insert a blank line below the line the cursor is on.
:set number	Show line numbers.